

Khaja Afreed Shaik

(224) 464 5334 | afreedkhaja12@gmail.com | <https://in.linkedin.com/in/khaja-afreed-shaik>
<https://github.com/afreedkhaja12-creator>

SUMMARY

Experienced Machine Learning Engineer/AI Engineer with **4+ years of experience** in designing highly scalable, high-performance Machine Learning and AI solutions. Proven track record in optimizing ML/data pipelines and reducing processing times. Proficient in Python, SQL, JAVA, Azure ML, Snowflake, MLflow, DBT and cloud-based data processing, with a strong background in MLOps and generative AI projects. Ready to leverage technical expertise and leadership skills to drive innovation and enhance data-driven decision-making.

PROFESSIONAL EXPERIENCE

Infosys Limited – Specialist Programmer

Aug 2022 – Present

MLOps & Scalable Pipelines

- **Architected and scaled enterprise-grade ML pipelines** using **Azure ML Studio** for **demand forecasting, sales and marketing** domains
- Deployed and monitored more than **35+ pipelines** across international markets using **Ray , Pyspark & Databricks** distributed computing (**5–40 node clusters**), reducing forecast generation time down by **75%** with **80%+ accuracy** across a **multiple million \$'s portfolio**
- Implemented **A/B testing, MLflow** Artifacts registration, continuous monitoring by building **Aporia Dashboards**, and ensured compliance with data governance using **Great Expectations**, for production ML models to ensure robust and reliable performance
- Collaborated with cross-functional teams to align project goals, manage backlogs, and refactor data science codebases to production standards, resulting in a **25% improvement in decision-making speed**
- Implemented **MLOps CI/CD pipelines** using **Azure Pipelines** to automate **Pytest, vulnerability scans, Docker image builds**, and ML endpoint submission to **Azure ML Workspace**, securing credentials via **Azure Key Vault**
- Leveraged **PySpark, Databricks, and Ray** for advanced **parallel processing** within ML pipelines, successfully **reducing pipeline runtime by 80%** and significantly enhancing model scalability

Advanced Machine Learning Solutions

- **Developed meta-learning ensemble system** with automated tournament-based **model selection** across **12+ ML algorithms (LightGBM, XGBoost, CatBoost, Random Forest)**
- **This system** incorporates analysis and dynamic winner identification based on model performance which Improved forecast accuracy by **30%** across hierarchical levels of the products
- **Collaboratively developed a modular Python library for end-to-end time-series forecasting**, encapsulating data retrieval, preprocessing, feature engineering, modeling, post-processing, and automated data uploads
- Enabled **60–75% code reuse** and **reduced build time by 70%**, accelerating new POC development from **weeks to days** and saving an estimated **\$15K–\$25K** per project in engineering effort and infrastructure costs

AI & Generative AI Projects

- Engineered scalable **AI** applications and **FastAPI** APIs using a **Multi-Container Project (MCP) server** for high-throughput, low-latency LLM serving
- Developed and deployed a **RAG-based Document Search API** and a **Text-to-SQL chatbot**, leveraging **LangChain, Langflow, and Hugging Face, Kubernetes** for natural language query capabilities over structured and unstructured data
- Fine-tuned and deployed LLM models from **OpenAI, Cohere, and Claude**, optimizing inference pipelines to improve accuracy and serving efficiency

- Implemented **vector embeddings** and integrated with vector datastores like **Pinecone and FAISS** to enable efficient semantic search and retrieval for RAG applications

Infosys Limited – System Engineer

Aug 2021 – Aug 2022

Data Engineering & Architecture

- Designed and implemented **ELT Pipelines** using **DBT** and **Dagster** to orchestrate automated data movement of **5TB+ daily**, providing valuable business insights
- Built a modern data architecture with **Python, DBT, Dagster, and Snowflake**, processing over **10M+ daily transactions** to drive actionable insights
- Implemented query optimization techniques on **75+ frequently run queries**, resulting in a **30% reduction in Snowflake credits consumption** and an estimated saving of **\$15k** in monthly cloud costs
- Ensured compliance with data governance using **Great Expectations**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL.
- **Python libraries:** Pandas, Numpy, PySpark, Matplotlib, NLTK, Pytorch, Scikit-learn, LangChain, MLflow, SciPy, Langflow.
- **ML Frameworks:** LightGBM, XGBoost, CatBoost, Random Forest, Scikit-learn
- **Data Infrastructure:** Snowflake, Azure Blob storage, AWS S3, MongoDB, Pinecone, Snowflake, FAISS
- **ETL/ELT frameworks:** Dagester, Databricks, DBT
- **Additional Tools:** MS Excel, Git, Jenkins, Confluence, CI/CD, Azure DevOps, Miro, Docker, Terraform
- **Statistical Techniques:** Linear Regression, Logistic Regression, Decision Tree, A/B Testing, Hyper-parameter tuning

EDUCATION

Qis college of engineering and Technology

India | Apr 2021

- **Bachelor of Technology** Computer science and Engineering

CERTIFICATIONS

- AI-102: Azure AI Engineer Associate
- Infosys Certified Big Data Developer
- Infosys Certified AI Master
- Infosys Certified Python Developer
- Infosys Certified Professional - Generative AI for All